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Original Study

Association of Racial and Ethnic Composition With Staff Levels in Nursing Homes: 2013-2019

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A B S T R A C T

Objectives: The nursing home (NH) population has become increasingly diverse, yet many facilities remain de facto racially segregated. This study examines whether a high proportion of Black, Indigenous, and People of Color (BIPOC) residents is associated with nursing staff levels.
Design: We constructed a longitudinal cohort of NHs (2013-2019) by linking Certification and Survey Provider Enhanced Reports, LTCFocUS.org, Medicare Cost Reports, and Payroll-Based Journal data. Separate multivariable random effects linear regressions were conducted.
Setting and Participants: 14,075 Medicare- and Medicaid-certified NHs in the United States.
Methods: The proportion of BIPOC residents was categorized as the 10% of nursing homes serving the highest minority residents in each state each year (High-BIPOC) and the remaining 90% (Low-BIPOC). Total nursing staff levels in hours per resident-day (HPRD) included both hours paid (2013-2019) and hours worked (2017-2019). The total staff included registered nurses, licensed practical nurses, and certified nurse aides.
Results: The unadjusted difference in total staff levels between High-BIPOC and Low-BIPOC NHs increased from −0.23 HPRD (4.19 vs 4.42) in 2013 to −0.35 HPRD (3.94 vs 4.29) in 2019 for hours paid. The difference in hours worked increased from −0.19 (3.55 vs 3.74) in 2017 to −0.23 (3.50 vs 3.73) in 2019. The difference became smaller but remained significant after controlling for covariates (−0.037 HPRD for hours paid, and −0.038 for hours worked). Analyses of individual staff types found lower levels of registered nurses and certified nurse aides (but not licensed practical nurses) among High-BIPOC nursing homes. Findings were robust to treating racial and ethnic composition as a continuous variable or excluding payer mix from the models.
Conclusions and Implications: NHs with high concentrations of minority residents reported lower nursing staff levels. Improving staffing in NHs serving primarily marginalized racial and ethnic groups remains a policy priority.
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Over the past two decades, nursing homes have served a more racially and ethnically diverse population. From 1999 to 2008, the proportion of Blacks (from 9.6% to 11.3%), Hispanics (from 2.3% to 3.8%), and Asians (from 1.0% to 1.6%) increased, whereas the proportion of non-Hispanic White residents decreased from 86.7% to 82.9%.[1] This demographic shift continued between 2011 and 2017.[2] The growing number of Black, Indigenous, and People of Color (BIPOC) residents, however, has not changed the highly segregated nature of

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nursing home care.[3] Approximately 50% of BIPOC residents reside in only 10% of US nursing homes.[4-7]

Compared to White residents, newly admitted Black and Hispanic residents report higher levels of functional and cognitive impairments,[8] leading to a greater need for nursing and personal care. Despite this increasing need, nursing homes with high concentrations of BIPOC residents tend to have fewer available financial resources.[4] The percentage of Medicaid residents was 57% in facilities with <5% BIPOC residents vs 77% in facilities with ≥35% BIPOC residents.[9] Medicaid reimburses nursing homes at lower rates than Medicare or private payers.[10] A greater care need coupled with fewer financial resources makes it harder for minority-serving nursing homes to recruit and retain nursing staff, resulting in racial and ethnic disparities in staffing. Inadequate staff levels could compromise nursing homes' quality of care[7,9,11-13] and hurt residents' quality of life.[14]

Empirical evidence suggests that nursing homes with high concentrations of BIPOC residents have lower staffing levels.[2,9] These studies, however, often rely on data sources like Certification and Survey Provider Enhanced Reports (CASPER) that collected self-reported staffing information from nursing homes.[15] Since the early 2010s, the Centers for Medicare and Medicaid Services (CMS) started collecting more reliable staffing data based on accounting information in the Medicare Cost Reports[16] and, in 2017, employee payroll records in the Payroll-Based Journal (PBJ) system.[17] No study has examined differences in staffing levels by racial and ethnic compositions using longitudinal Medicare Cost Reports or PBJ data.

To mitigate these knowledge gaps, we used 2013-2019 national nursing home data to describe the time trends in nursing staff levels in nursing homes with high vs low concentrations of racial and ethnic minority residents. We then performed separate longitudinal regressions to examine the associations of a facility's racial and ethnic composition with total and individual nursing staff levels.

Methods

Data Sources

We linked the 2013-2019 CASPER data with the LTCFocUS.org data and Medicare Cost Reports, all at the facility level. This study focused on data from 2013 to 2019 to reflect normal operations of nursing homes before the COVID-19 pandemic. The CASPER data were collected as part of initial and annual recertification inspections of all Medicare- and Medicaid-certified nursing homes, with these inspections occurring every 9-15 months.[18,19] CASPER included detailed facility characteristics, such as chain affiliation, ownership, and payer mix.[20] The LTCFocUS.org data provided annual summarized information on nursing homes from CASPER and Minimum Data Set assessments.[2] The Medicare Cost Reports were mandatory annual cost reports for Medicare-certified facilities, which collected the number of hours paid to nursing staff.[16] We also obtained the 2017-2019 PBJ data for the number of hours worked by nursing staff based on auditable payroll records.[17] PBJ data were unavailable before 2017. Cost Reports and PBJ data provided the best available data on nursing home staffing.[16,20,21] The 2010 Rural-Urban Commuting Area Codes identified the rurality of nursing homes.[22] This study was exempted by the University of XXX Institutional Review Board. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cohort studies.

Study Cohort

We assembled a longitudinal cohort of all Medicare- and Medicaid-certified nursing homes in the United States for 2013-2019 (Supplementary Figure 1). We initially identified 16,107 unique nursing homes that completed the CASPER surveys. We excluded 1150 nursing homes that could not be linked to the Medicare Cost Reports, most of which were hospital-based facilities. We then dropped 837 nursing homes without information on the racial and ethnic composition of residents. Finally, we excluded 45 nursing homes with aberrant staffing-level data (ie, zero staff, beyond ±4 SDs, or >24 HPRD).[16,20,21] This resulted in a final longitudinal cohort of 84,722 facility-year observations from 14,075 unique nursing homes, with each facility having up to 7 observations from 2013 to 2019.[20]

Exposure and Measures

The independent variable is the racial and ethnic composition of residents in a facility.[2] LTCFocUS.org calculated the composition among all residents who resided in a nursing home on the first Thursday in April using Minimum Data Set assessments.[2] BIPOC residents included those who were Blacks, Hispanics, Asians, Native Americans, and Pacific Islanders, excluding non-Hispanic Whites.[23,24] In the primary analyses, we ranked nursing homes based on the proportion of BIPOC residents in each state for each year. Our BIPOC concentration variable defined "High-BIPOC" as the top 10% of nursing homes serving the largest proportion of BIPOC residents in each state for each year.[25] The bottom 90% of nursing homes were classified as "Low-BIPOC." This approach acknowledged the fact that both racial and ethnic compositions in nursing homes and state staffing regulations vary substantially by state.[24,26] By using state-specific cutoffs, we aimed to understand how the most racially and ethnically diverse nursing homes in each state operate relative to other nursing homes within the same state.[24,27] A sensitivity analysis using the top 25% as the cutoff produced similar results.

Staffing outcomes included 2 measures of total nursing staff levels in hours per resident-day (HPRD): hours paid (2013-2019) and hours worked (2017-2019).[16] The total nursing staff included registered nurses (RNs), licensed practical nurses (LPNs), and certified nursing assistants (CNAs). RNs provide comprehensive patient care, coordinate treatment plans, and supervise other nursing staff.[28] LPNs provide nursing care under the supervision of an RN, who along with CNAs provide direct care in support of daily living activities, like bathing and feeding.[29,30] The hours paid from Medicare Cost Reports included hours for direct care and meal breaks, paid time off, and vacation time. The hours worked from PBJ excluded any time (paid or unpaid) for meal breaks, paid time off, and vacation time.[16] Hours paid was available for the whole study period. Hours worked, although only available since 2017 in the PBJ data, better reflects the number of hours staff care for residents.

We controlled for nursing home covariates potentially associated with the staffing outcomes: chain affiliation, ownership, number of beds, occupancy rate, case-mix acuity index, percentage of residents with dementia, percentage of residents with depression, percentage of residents with serious mental illness, percentage of Medicaid residents, percentage of Medicare residents, and rurality.[5,12,14,31-33]

Statistical Analysis

We first described the variations in the proportion of BIPOC residents over time and by facility. We then presented the cutoff values of the top 10% in each state in 2019. We tested whether nursing home characteristics in 2019 differed by BIPOC concentration using *t* tests for continuous variables and χ^2 tests for categorical variables. We plotted the time trends in the 2 staffing measures (hours paid 2013-2019, and hours worked 2017-2019) by High- vs Low-BIPOC concentration. We also assessed the bivariate relationship of racial and ethnic composition with 2 staffing measures using 2019 data.

To examine the longitudinal associations of BIPOC concentration with staffing outcomes, we employed separate panel random effects linear regressions for each outcome. The random effect models

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accounted for the correlation within the same nursing home and the variability between nursing homes. Fixed effects regressions were deemed inappropriate because the independent variable, BIPOC concentration, varied minimally within a facility over time.[34] Each nursing home had a maximum of 7 observations for the total nursing staff level (hours paid) from 2013 to 2019 and had a maximum of 3 observations for the total nursing staff level (hours worked) from 2017 to 2019. These models adjusted for year fixed effects, state fixed effects, and the facility covariates described previously. We also examined potential heterogenous associations of BIPOC concentration with different types of staff, by repeating the random effects linear regressions on individual staffing levels of RNs, LPNs, and CNAs for hours paid and hours worked.

We performed 2 sets of sensitivity analyses. To assess the robustness of our primary findings, the first set of analyses repeated the longitudinal models on total staff levels with the continuous racial and ethnic composition variable. Because nursing homes caring for BIPOC residents relied more on Medicaid payments, the second set of analyses examined the impact of removing payer mix on our findings by estimating the longitudinal models without adjusting for the percentage of Medicare residents and the percentage of Medicaid residents.

All statistical analyses were performed using SAS 9.4 (SAS Institute Inc) and Stata 16.0 (StataCorp LLC).

Results

The proportion of BIPOC residents in nursing homes remained stable from 2013 to 2019, ranging from 11.38% to 11.73%. There were large variations in the proportion across facilities (Supplementary Figure 2). The cutoff values for the top 10% of nursing homes serving the largest proportion of BIPOC residents varied substantially across states, ranging from 6% of BIPOC residents to >70% (Supplementary Table 1). Table 1 compares the characteristics of the top 10% of nursing homes vs the bottom 90% of facilities in 2019. High-BIPOC facilities were more likely to be for-profit (85.8% vs 72.1%), had more beds (127.8 beds vs 109.9 beds), had a higher percentage of Medicaid residents (70.5% vs 59.6%), and were more likely located in urban areas (95.4% vs 89.8%). In the bivariate analyses, racial and ethnic composition was negatively associated with total nursing staff levels (Supplementary Figure 3), with a Pearson correlation coefficient of −0.12 for hours paid and −0.16 for hours worked.

Figure 1 examines the trends in total nursing staff-hours per resident-day (hours paid and hours worked) in nursing homes by BIPOC concentration from 2013 to 2019. Overall, High-BIPOC nursing homes reported lower staffing levels (hours paid) than Low-BIPOC nursing homes. In 2013, staffing levels (hours paid) were 4.19 HPRD in High-BIPOC nursing homes vs 4.42 HPRD in Low-BIPOC nursing homes. In 2019, total staffing levels (hours paid) decreased to 3.94 HPRD in High-BIPOC nursing homes and to 4.29 HPRD in Low-BIPOC nursing homes. Thus, the difference increased from 2013 to 2019 (−0.23 HPRD vs −0.35 HPRD). From 2017 to 2019, the difference in total staffing levels (hours worked) slightly increased from −0.19 HPRD in 2017 to −0.23 HPRD.

To assess the association of BIPOC status with total staffing levels, we performed two separate random effects linear regressions on hours paid using 2013-2019 Medicare Cost Report data and on hours worked using 2017-2019 PBJ data (Table 2). Compared with Low-BIPOC nursing homes, High-BIPOC nursing homes reported lower hours paid ($\beta = -0.037$ HPRD, 95% CI: −0.08, −0.004) and lower hours worked for total nursing staff ($\beta = -0.038$ HPRD, 95% CI: −0.06, −0.02). In the analyses of individual staff levels, High-BIPOC nursing homes reported lower hours paid for RNs ($\beta = -0.02$ HPRD, 95% CI: −0.03, −0.001) and CNAs ($\beta = -0.03$ HPRD, 95% CI: −0.05, −0.01) (Table 3), but not LPNs. Similar findings on hours worked were found in Table 3.

Finally, sensitivity analyses confirmed the negative association of racial and ethnic composition with total nursing staff levels when using the continuous BIPOC variable (Supplementary Table 2). Another set of sensitivity analyses examined whether removing payer mix from the random effects regressions affected the association of BIPOC status with staff levels (Supplementary Table 3). Overall, removing payer mix resulted in minimum changes in the estimates for total and individual staff levels (both hours paid and hours worked).

Table 1
Nursing Home Characteristics by Racial and Ethnic Composition, 2019

Variables	All Nursing Homes (N = 11,960)	High-BIPOC Nursing Homes* (n = 1225)	Low-BIPOC Nursing Homes† (n = 10,735)	*P* Value‡
Nursing home characteristics				
Chain affiliation, n (%)	7203 (60.23)	741 (60.49)	6462 (60.20)	.84
Ownership, n (%)				
For-profit	8791 (73.50)	1051 (85.80)	7740 (72.10)	<.001
Government	592 (4.95)	36 (2.94)	556 (5.18)	
Not-for-profit	2577 (21.55)	138 (11.27)	2439 (22.72)	
Number of beds	111.73 ± 56.76	127.8 ± 63.99	109.9 ± 55.58	<.001
Occupancy rate (0-100)	79.19 ± 15.43	79.90 ± 14.38	79.12 ± 15.55	.07
Case-mix acuity index	10.45 ± 1.26	10.71 ± 1.46	10.42 ± 1.2923	<.001
% residents with dementia	44.85 ± 16.25	41.22 ± 16.22	45.27 ± 16.21	<.001
% residents with serious mental illness	33.95 ± 17.74	34.31 ± 17.66	33.91 ± 17.75	.45
% residents with depression	36.58 ± 23.24	31.24 ± 22.41	37.19 ± 23.26	<.001
% Medicare residents (0-100)	12.48 ± 11.18	10.40 ± 9.29	12.72 ± 11.35	<.001
% Medicaid residents (0-100)	60.73 ± 22.08	70.48 ± 19.22	59.62 ± 22.11	<.001
Rurality, n (%)				
Urban	10,804 (90.33)	1168 (95.35)	9636 (89.76)	<.001
Rural	1156 (9.67)	57 (4.65)	1099 (10.24)	
Outcome measures				
Total nursing staff-hours per resident-day (hours worked)	3.73 ± 0.70	3.54 ± 0.63	3.75 ± 0.71	<.001
Total nursing staff-hours per resident-day (hours paid)	4.36 ± 1.61	4.07 ± 1.41	4.39 ± 1.63	<.001

Unless otherwise noted, values are mean ± SD.
*High-BIPOC nursing homes are defined as the top 10% of nursing homes serving the largest proportion of minority residents in each state.
†The bottom 90% of nursing homes are classified as Low-BIPOC.
‡*t* tests for continuous variables, χ^2 tests for categorical variables.

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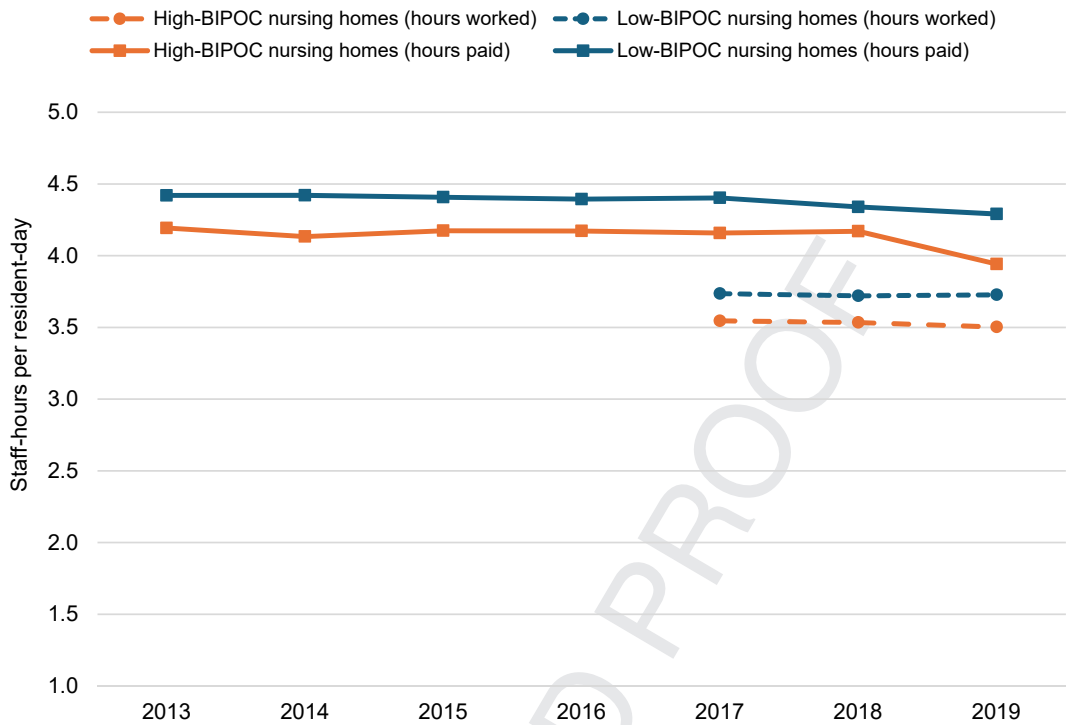

Fig. 1. Trends in total nursing staff levels in nursing homes by racial and ethnic composition, 2013-2019. The nursing staff levels are measured in hours per resident-day. Data on hours paid for 2013-2019 are from the Medicare Cost Reports; data on hours worked for 2017-2019 are from the Payroll-Based Journal (PBJ). The number of nursing homes included each year ranges from 11,834 to 12,302. High-BIPOC is defined as the top 10% of facilities serving the largest proportion of Black, Indigenous, and People of Color (BIPOC) residents in each state for each year. The bottom 90% of nursing homes are classified as Low-BIPOC. The 95% CIs are not presented because of small variations.

Discussion

Using longitudinal Medicare Cost Reports and PBJ data, this study found that nursing homes with a higher concentration of racial and ethnic minority residents consistently reported lower nursing staff levels. In the longitudinal models, disparities in nursing staff levels became smaller but remained significant after controlling facility characteristics. These results are robust to the staffing measures used (hours paid or hours worked), the measures of racial and ethnic composition (bivariate or continuous), or whether the payer mix was controlled or not. Our findings are consistent with previous studies using CASPER data or cross-sectional PBJ data, providing further evidence of racial and ethnic disparities in nursing home staffing.[2,9]

Table 2
Random Effects Regressions Examining the Association of Racial and Ethnic Composition With Total Nursing Staff Levels

Variables	Total Nursing Staff-Hours per Resident-Day (Hours Paid), 2013-2019*	Total Nursing Staff-Hours per Resident Day (Hours Worked), 2017-2019†
Independent variable: High-BIPOC (vs low-BIPOC)	−0.037 (−0.079 to −0.017)	−0.038 (−0.058 to −0.017)
Nursing home characteristics		
Chain affiliation	−0.080 (−0.102 to −0.058)	−0.084 (−0.098 to −0.071)
Ownership (Ref: For-profit)		
Government	0.149 (0.097 to 0.200)	0.239 (0.208 to 0.269)
Not-for-profit	0.329 (0.291 to 0.366)	0.311 (0.292 to 0.330)
Number of beds	−0.0001 (−0.001 to 0)	−0.001 (−0.001 to −0.001)
Occupancy rate (0-100)	−0.011 (−0.012 to −0.010)	−0.007 (−0.007 to −0.006)
Case-mix acuity index	0.009 (0.001 to 0.017)	0.039 (0.034 to 0.044)
% residents with dementia	−0.00004 (−0.001 to 0.001)	0.001 (0.001 to 0.001)
% residents with serious mental illness	−0.001 (−0.001 to 0)	−0.002 (−0.002 to −0.001)
% residents with depression	0.0001 (−0.000 to 0.001)	0.0002 (−0.000 to 0)
% Medicare residents (0-100)	0.008 (0.006 to 0.0081)	0.007 (0.006 to 0.007)
% Medicaid residents (0-100)	−0.004 (−0.005 to −0.003)	−0.007 (−0.007 to −0.006)
Rural (Ref: Urban)	−0.16 (−0.245 to −0.075)	−0.066 (−0.095 to −0.036)
Observations, n	84,722	35,651
Unique nursing homes, n	14,075	13,649

High-BIPOC, the top 10% of facilities serving the largest proportion of BIPOC residents in each state for each year; Low-BIPOC, the bottom 90% of nursing homes in each state for each year.
Unless otherwise noted, data are expressed as coefficient (95% CI). Year and state fixed effects are not presented.
*Data on hours paid for 2013-2019 are from the Medicare Cost Reports, with each nursing home having up to 7 facility-year observations.
†Data on hours worked for 2017-2019 are from the Payroll-Based Journal (PBJ), with each nursing home having up to 3 facility-year observations.

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Table 3
Random Effects Regressions Examining the Association of Racial and Ethnic Composition With Individual Nursing Staff Levels

Variables	Staff-Hours per Resident-Day (Hours Paid), 2013-2019*			Staff-Hours per Resident-Day (Hours Worked), 2017-2019†		
	Registered Nurse	Licensed Practical Nurse	Certified Nursing Assistant	Registered Nurse	Licensed Practical Nurse	Certified Nursing Assistant
Independent variable: High-BIPOC (vs low-BIPOC)‡	−0.018 (−0.029 to −0.001)	0.009 (−0.001 to 0.019)	−0.027 (−0.049 to −0.005)	−0.020 (−0.028 to −0.012)	0.007 (−0.001 to 0.016)	−0.021 (−0.036 to −0.006)
Observations, n	84,722	84,722	84,722	35,951	35,951	35,951
Unique Nursing homes, n	14,075	14,075	14,075	13,649	13,649	13,649

High-BIPOC, the top 10% of facilities serving the largest proportion of BIPOC residents in each state for each year; Low-BIPOC, the bottom 90% of nursing homes in each state for each year.
Covariates (chain affiliation, ownership, number of beds, occupancy rate, case-mix acuity index, % residents with dementia, % residents with serious mental illness, % residents with depression, and rurality), as well as year and state fixed effects are not presented.
*Data on hours paid for 2013-2019 are from the Medicare Cost Reports, with each nursing home having up to 7 facility-year observations.
†Data on hours worked for 2017-2019 are from the Payroll-Based Journal (PBJ), with each nursing home having up to 3 facility-year observations.
‡Data are expressed as coefficient (95% CI).

One concerning finding is that disparities in nursing staff levels (both hours paid and hours worked) worsened in 2019, before the COVID-19 pandemic hit nursing homes. Nursing homes lost 220,000 employees in 2020[35,36] and lag hospitals and home health agencies in recovering their staff.[18,37] More than 20% of facilities still reported shortages in nursing staff and aides in September 2022.[38] Almost half of nursing homes now rely on costly agency/contract workers to fill the staffing gaps.[21] This can especially hurt minority-serving nursing homes with limited financial resources. Therefore, disparities in staffing levels may become worse during and after the pandemic.[13,18,39]

The analyses of individual types of nursing staff found lower levels of RNs and CNAs among nursing homes with high concentrations of BIPOC residents. Staff levels for LPNs did not differ by racial and ethnic composition. Nursing homes may use LPNs to substitute for more skilled RNs, which could lower the quality of care being delivered as LPNs practice near the frontiers of their scope of practice.[28] Lower levels of CNAs in High-BIPOC facilities may reduce direct personal care provided to residents. Disparities in RN and CNA levels may change in the coming years because of the federal staffing mandates. In May 2024, CMS finalized the rule for the first-ever federal minimum staffing standards for nursing homes.[40,41] The proposal included a minimum of 0.55 HPRD for RNs, 2.45 HPRD for CNAs, and 3.48 HPRD for total nursing staff. Many nursing homes would hire more staff or use agency staff to fulfill the new requirements.[18,21,42] This could lead to reduced disparities in nursing staff levels because High-BIPOC nursing homes have to increase their staff levels more to meet the mandates. For example, in 2019, we found that only 40% of nursing homes with high concentrations of BIPOC residents would have met the RN standard vs 53% of other facilities. Despite the promises, policy makers should ensure that minority-serving facilities have the necessary resources to recruit additional staff.[43]

Reliance on Medicaid payments could have explained the lower staffing levels in nursing homes with a high concentration of BIPOC residents.[32] However, excluding percentage Medicaid residents and percentage Medicare residents did not change the direction of the association of BIPOC status with staff levels. This could be due to the inclusion of a random effect, or this may suggest that other structural factors beyond financial resources also contribute to poorer staffing levels in High-BIPOC nursing homes.[25] For example, High-BIPOC nursing homes are more likely to be for-profit, which is associated with lower staffing.[44] Also, nursing homes with mostly BIPOC residents may have poorer working environments that hurt staff levels.[28,45]

The fundamental question is how can we minimize (if not eliminate) the enduring racial and ethnic disparities in nursing home staffing? A solution would be to address systemic racism in the design of health care systems, which leads to racial and spatial segregation[3,5,6,8,12,14,25,46] and makes it harder for historically marginalized groups to access high-quality nursing homes.[2,8] Our health system and policy makers should reduce barriers (geographic, financial, and informational) faced by historically marginalized older adults to access high-quality facilities. Because nursing homes serving these populations are more likely to be located in severely deprived neighborhoods, policies such as payment enhancement and workforce development grants that could be micro-targeted to those facilities should improve staff levels.[46] Practice initiatives may also strengthen staff recruitment, training, and retention in nursing homes with high concentrations of BIPOC residents.[8,25,47] One such example is to improve prelicensure training experiences in nursing homes through academic and practice partnerships.[48] Programs that improve the organizational culture and working environment should also increase retention and reduce turnover of nursing staff.[49] Culturally specific programs in nursing homes with high concentrations of BIPOC residents may also help. Those programs can improve nursing staff's

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competency in meeting the needs of diverse residents, hence reducing staff burnout and turnover.[47]

Limitations

Staffing measures may have errors, including facilities reporting zero staff, >24 staff-hours per resident-day, and beyond ±4 SDs.[16] These aberrant staff levels are theoretically impossible for nursing homes to operate, and hence are likely due to errors.[16,20,21] Excluding these facilities should have minimized the impact of aberrant staffing levels on our analyses. We also did not examine differences in other aspects of nursing home staffing, such as turnover and staff instability. Because of the observational study design, we cannot ascertain a causality of racial and ethnic composition with staffing. The year and state fixed effects in the models removed the impact of time trends and variations in state policies, providing conservative estimates. Finally, the underlying mechanisms leading to racial and ethnic disparities need to be further examined.

Conclusions and Implications

Using longitudinal data from 2013 to 2019, we found persistent racial and ethnic disparities in nursing home staff levels. The tolls of racial and ethnic segregation in nursing homes cannot be ignored. Staff shortages during the COVID-19 pandemic may have further enlarged existing racial and ethnic disparities in staffing.[13,18,21,50] Additional research is needed to identify policy and practice initiatives that can mitigate nursing home disparities, including understanding the role of facility culture and working environment in contributing to disparities in staff levels.

Disclosure

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Supplementary Data

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